

Assignment 4: Generative Models with GAN

Ben-Gurion University of the Negev - Deep Learning Course

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1. Introduction

This report presents a standard GAN and Conditional GAN (cGAN) trained on the Adult Census Income dataset for tabular data synthesis, together with: a comparison of softmax vs Gumbel-ST discrete representations (Section 6), a deliberate mode-collapse experiment with minibatch-std mitigation (Section 7), and a spectral-normalization contribution (Section 8).

2. Dataset and Preprocessing

The Adult dataset (32,561 records, 1994 US Census) has a binary income target (75.9% $\leq 50K$ / 24.1% $> 50K$, Fig. A1). Six continuous features are standardized with StandardScaler (not MinMaxScaler: the linear output head is unbounded). Eight categorical features are one-hot encoded (not ordinal: ordinal encoding implies invalid linear interpolation between levels), yielding a 108-dimensional input. Missing values in workclass/occupation/native-country are mapped to "Unknown" rather than imputed (mode imputation corrupts the target distribution) or dropped (survivorship bias). Feature distributions are in Appendix Figs. A2–A3.

Train/test split: 80% training (26,048 records) and 20% test (6,513 records), stratified on the income label to preserve the 0.759/0.241 class ratio. All results are averaged over three random seeds (42, 123, 456).

3. Model Architecture

3.1 Generator

The generator is a three-layer MLP with 256 hidden units per layer, batch normalization, and LeakyReLU(0.2) activations. Three layers of 256 units were chosen to give the network sufficient capacity to model the 108-dimensional encoded space while remaining trainable without skip connections; shallower networks underfit the joint categorical and continuous structure, and wider layers produced no measurable gain. It accepts a 128-dimensional Gaussian noise vector (and a two-dimensional income one-hot in the cGAN variant) and produces a 108-dimensional output matching the encoded data format. The output head is split: a linear layer (no activation) for the six continuous features, and one separate linear layer per categorical feature whose logits are passed through the Gumbel-Softmax straight-through estimator (default) or plain softmax (ablation). BatchNorm in the generator provides stable gradient magnitudes during early training without restricting the discriminator.

3.2 Discriminator

The discriminator is a three-layer MLP with 256, 256, and 128 hidden units, LeakyReLU(0.2) activations, and Dropout(0.3) after the first two layers. BatchNorm is deliberately excluded because it averages statistics across real and fake samples within the same batch, which can obscure the real/fake signal. The output is a single sigmoid unit. For the cGAN, the income one-hot is concatenated to the input so the discriminator learns to assess plausibility conditional on the requested class.

3.3 cGAN Extension

The income one-hot (dim 2) is concatenated to the noise vector (G input) and to the data vector (D input), so both networks see the same conditioning signal and D always evaluates plausibility given the requested class.

4. Training Setup

Hyperparameter	Value
Epochs	200 (Sections 6/9); 100 (Section 7); 80 (Section 8)
Batch size	256
Noise dimension z	128
Optimizer	Adam, $lr = 2e-4$, $b1 = 0.5$, $b2 = 0.999$
Loss function	Binary cross-entropy (BCE)
Label smoothing	real $\rightarrow 0.9$, fake $\rightarrow 0.1$ (two-sided)
Temperature schedule	Linear anneal 1.0 $\rightarrow$ 0.5 over training
D : G update ratio	1:1 (default); 5:1, hidden_g=16, softmax (Section 8 collapse)
Categorical strategy	Gumbel-ST (default); Softmax (Section 7 ablation)

Two-sided label smoothing is used: real samples are assigned a target of 0.9 (instead of 1.0) and fake samples a target of 0.1 (instead of 0.0). Smoothing the real target prevents the discriminator from becoming overconfident and producing near-zero gradients for the generator. Smoothing the fake target provides a symmetric regularisation effect, preventing the discriminator from driving its fake-sample output to exactly 0. The Gumbel-Softmax temperature is annealed from 1.0 to 0.5 so early training benefits from softer, more gradient-rich outputs while later training commits to near-discrete samples.

5. Evaluation (Section 6)

5.1 Loss Curves

Loss curves for all three seeds are shown in Appendix Figs. A4-A9. In all runs the generator loss converges to approximately 2.27-2.29 and the discriminator loss to approximately 0.65-0.66. With two-sided label smoothing (real label = 0.9, fake label = 0.1), the discriminator's optimal output is $D(\text{real}) = 0.9$ and $D(\text{fake}) = 0.1$, yielding a minimum BCE loss of $\text{BCE}(0.9, 0.9) + \text{BCE}(0.1, 0.1) \approx 0.325 + 0.325 = 0.65$. This is the discriminator-dominance plateau - D has converged to perfectly classify real from fake at the smoothed targets - not the game-theoretic Nash equilibrium, which requires G to match p_{data} so that D outputs 0.5 for all samples, giving D-loss $\approx \text{BCE}(0.5, 0.9) + \text{BCE}(0.5, 0.1) = 0.693 + 0.693 \approx 1.386$ (as confirmed by the spectral-normalization experiment in Section 8). The generator loss well above $\ln(2) = 0.693$ confirms the discriminator retains a consistent advantage throughout training. Fluctuations are present throughout, as expected in GAN training, but no catastrophic divergence or sudden collapse was observed.

5.2 Synthetic Data Quality

Continuous and categorical feature distributions for both GAN and cGAN are compared against real training data in Appendix Figs. A10-A13. The continuous distributions are qualitatively similar in shape, though the synthetic capital-gain and capital-loss distributions lack the heavy-tail structure of the real data; these features are zero-inflated (>90% of values are exactly 0) with a sparse heavy tail, a degenerate manifold that Gaussian noise mapped through a smooth MLP cannot naturally reproduce without explicit zero-mass modeling. Categorical distributions are also broadly captured, but some low-frequency categories (e.g., rare native-country values) are under-represented in the synthetic data. Correlation matrices (Figs. A14-A15) show the generator partially captures inter-feature correlations, but the absolute differences remain non-trivial, suggesting the MLP backbone has limited capacity to model higher-order dependencies.

5.3 Detection and Efficacy Metrics

Detection is measured by training a Random Forest on a balanced mix of real and synthetic data using four-fold cross-validation (three folds train, one fold test, averaged over four rotations) and reporting the AUC. Low AUC near 0.5 indicates synthetic data is indistinguishable from real. Efficacy is the ratio of AUC when a Random Forest is trained on synthetic versus real data and evaluated on the original test set; high values near 1.0 indicate the synthetic data is a useful substitute for training.

Model	Detection AUC	Efficacy (mean)	Efficacy (std)
GAN	1.000 +/- 0.000	0.842	0.084
cGAN	1.000 +/- 0.000	0.763	0.105

Detection AUC is 1.000 for both models across all three seeds, meaning the Random Forest perfectly separates real from synthetic samples with no overlap. When D converges to its smoothed-label optimum while G-loss stays well above $\ln(2)=0.693$, the generator has not matched p_{data} and the synthetic manifold is detectably different from the real one.

Consistency across all three seeds rules out initialization as the cause - this is a structural limitation of the MLP generator. Section 8 confirms that training stability is not the bottleneck either: even after spectral normalization drives training to the true Nash equilibrium, detection AUC remains at 1.000, isolating model capacity as the binding constraint.

GAN efficacy (mean 0.842 ± 0.084) shows that synthetic data trained a Random Forest to 84% of the performance achievable with real data - usable as a partial substitute. The cGAN is slightly lower (0.763 ± 0.105): conditioning on a prescribed class ratio constrains the generated distribution, while the unconditional GAN's pseudo-labeling naturally tracks the learned feature space. The high standard deviation across seeds for both models reflects sensitivity to the noise initialization, which the spectral-norm variant (Section 8) largely resolves (efficacy 0.992, std ≈ 0).

6. Discrete Feature Representation (Section 7)

6.1 Why Argmax Breaks Gradient Flow

Argmax is piecewise constant: for any small perturbation of the logit vector the output one-hot is unchanged, so the sub-gradient is zero almost everywhere. During backpropagation, the chain rule requires multiplying the downstream gradient by the local Jacobian of each operation; the Jacobian of argmax is zero, so the gradient signal reaching the generator's parameters is identically zero regardless of the discriminator's feedback. The generator therefore receives no information about which categorical values to produce and cannot improve its discrete outputs.

6.2 Strategy Comparison

Plain Softmax. Forward: the generator outputs a soft probability vector; the well-defined softmax Jacobian allows gradients to flow back to the logits. Backward: gradients are non-zero. However, the discriminator sees hard 0/1 one-hots from real data and soft probabilities from fake data - a format mismatch it can exploit instead of learning semantic content.

Gumbel-Softmax + Straight-Through (Gumbel-ST). Forward: Gumbel noise is added to the logits and a hard one-hot is taken via argmax, so the discriminator sees discrete samples identical in format to real data - no format mismatch. Backward: the straight-through estimator substitutes the soft Gumbel-Softmax gradient ($y_{\text{hard}} - y_{\text{soft}}.\text{detach}() + y_{\text{soft}}$) for the zero argmax gradient, allowing informative updates while the forward pass remains discrete.

6.3 Entropy Analysis

To quantify commitment, the average Shannon entropy of the generated soft categorical distributions (computed from raw logits, without Gumbel noise) is compared against the entropy implied by real category frequencies. A near-uniform distribution has high entropy and indicates the generator is not generating meaningful categorical structure. Results are shown in Appendix Fig. A16 and summarized below.

Feature	Real entropy	Softmax	Gumbel-ST
workclass	1.148	0.028	0.442
education	2.037	0.181	0.037
marital-status	1.273	0.058	0.505
occupation	2.440	0.095	1.070
relationship	1.491	0.070	0.664
race	0.551	0.010	0.448
sex	0.635	0.034	0.328
native-country	0.653	0.000	0.339

Softmax entropy is close to zero for every feature. The mechanistic reason is the format mismatch: real data arrives as hard 0/1 one-hot vectors, while Softmax fake data arrives as continuous probability vectors. The discriminator exploits this distributional difference rather than learning semantic content. The generator's best response is to concentrate probability mass on the single most frequent category per feature - driving logits large for the modal category and near-zero for all others - which produces peaked, low-entropy distributions (hence raw-logit entropy collapsing near zero). Gumbel-ST entropy is substantially closer to real-data entropy for most features (notably occupation: 1.070 vs real 2.440, far better than Softmax's 0.095). The one exception is education, where Gumbel-ST entropy (0.037) is lower than Softmax (0.181): education has the widest vocabulary (16 levels), and the annealed temperature ($1.0 \rightarrow 0.5$) pushes the straight-through estimator to commit aggressively to the single most frequent level. Softmax, paradoxically, does not collapse as far on this feature because the format-mismatch equilibrium keeps the discriminator focused on the soft/hard distinction rather than on the modal category. Loss curve comparison (Fig. A17) shows Gumbel-ST reaches a better equilibrium ($G \sim 2.29$, $D \sim 0.66$) than Softmax ($G \sim 1.88$, $D \sim 0.86$): Gumbel-ST forces the discriminator to learn genuine content differences, whereas Softmax's degenerate format-detection signal prevents D from converging to its optimum.

Quantitative comparison. Detection AUC: both 1.000 - the format mismatch does not help Softmax avoid detection, since the Random Forest finds other distinguishing features even

without the soft/hard signal. Efficacy: Softmax 0.944 vs Gumbel-ST 0.942 - nearly identical, suggesting that the soft probability vectors, despite being a distributional mismatch, still carry enough signal for a downstream classifier to learn useful structure. Despite the near-equal efficacy, Gumbel-ST is the correct representation: it eliminates the format mismatch, produces entropy values substantially closer to real-data category statistics, and reaches a theoretically sounder training equilibrium (D-loss closer to the label-smoothed optimum). Choosing Softmax because its efficacy is 0.002 higher would be overfitting to noise.

7. Mode Collapse: Induction and Mitigation (Section 8)

7.1 Collapse Indicator

The collapse indicator is the categorical coverage ratio: unique (workclass, marital-status, occupation) three-way combinations in synthetic data divided by those in real data. Values near 0 indicate collapse to a narrow pattern set; values near 1 indicate full diversity. These three features were chosen for their range of vocabulary sizes and social-demographic independence.

7.2 Inducing Collapse

Collapse is induced by combining: (1) a bottlenecked generator (hidden=16, down from 256), which cannot map noise to diverse categorical combinations; (2) plain softmax in place of Gumbel-ST, removing the diversity floor that Gumbel noise provides; and (3) a 5:1 D:G update ratio to maintain discriminator dominance.

Configuration	Coverage ratio	Unique combos
Real data (reference)	1.000	381
Baseline (1:1, hidden=256, Gumbel-ST)	1.031	393
Collapsed (5:1, hidden=16, softmax)	0.116	44

The revised strategy produced genuine mode collapse: coverage dropped from the baseline of 1.031 to 0.116, reducing unique (workclass, marital-status, occupation) combinations from 393 to just 44 out of 381 real combinations. The bottlenecked generator concentrated probability mass on the most frequent category for each feature, mapping nearly all noise vectors to the same few combinations. The loss curves (Fig. A18) reveal a key diagnostic property of mode collapse: despite the catastrophic diversity loss, the G-loss and D-loss plateaus are nearly identical to the baseline ($G \approx 2.27$, $D \approx 0.66$). This is because the discriminator already dominated in both cases, and scalar BCE loss is sensitive only to real/fake classification accuracy, not to output diversity. Mode collapse is therefore invisible in loss curves alone and can only be

detected through a coverage-based metric. This is why explicitly defining and tracking a collapse indicator (Section 7.1) is essential.

7.3 Mitigation: Minibatch Standard Deviation

Prediction. The minibatch standard-deviation (MBD) layer appends the mean per-feature standard deviation across the mini-batch to the discriminator input. When the collapsed generator produces near-identical fake samples, their batch std is near zero, making them trivially detectable. The generator must therefore spread its outputs to survive. Predicted: coverage ratio recovers above the collapsed value of 0.116; scalar losses remain similar (same architecture and update ratio).

Observed. MBD yielded a coverage ratio of 0.255 (97/381), more than double the collapsed run's 0.116 (44/381). G-loss settled at 2.30 and D-loss at 0.65 - nearly identical to the collapsed run, as predicted.

Prediction confirmed. MBD appends the mean per-feature standard deviation across the mini-batch to the discriminator's input. When the generator collapses, fake samples within a batch become nearly identical, driving their batch std toward zero - a signal the discriminator can exploit independently of whether any individual sample looks realistic. The generator is penalized not for a single fake sample but for producing a homogeneous batch, so it must spread its outputs across more combinations to survive. This explains the coverage increase from 44 to 97 unique combinations. Full recovery to the baseline (1.031) was not achieved because MBD improves diversity incentives but cannot expand representational capacity: with hidden=16, the generator lacks the parameters to span all 381 real combinations regardless of the gradient signal it receives. Bar chart and loss curves are in Appendix Figs. A19-A20.

8. Open-Ended Contribution: Spectral Normalization (Section 9)

8.1 Modification and Motivation

Spectral normalization (SN) is applied to every linear layer of the discriminator: each weight matrix W is divided by its largest singular value $\sigma_1(W)$ at each forward pass, bounding the discriminator's Lipschitz constant to 1. This targets the baseline instability where D converges quickly to a near-optimal solution, saturating its gradients (sigmoid output $\rightarrow 0$ for fake samples) and starving the generator of update signal. Unlike gradient penalty, SN requires no hyperparameter and adds no extra forward pass.

8.2 Prediction

Written before running the experiment. (1) G-loss variance should decrease: bounding the Lipschitz constant prevents sharp discriminator confidence spikes that cause the generator

gradient to oscillate. (2) D-loss should rise above the baseline plateau of ~ 0.65 : the constrained discriminator cannot converge to the smoothed-label optimum and should settle near the Nash value of ~ 1.386 . (3) Efficacy should improve: stable gradients give the generator a better learning signal. Detection AUC is not predicted to improve - if it does not, that indicates the bottleneck is model capacity rather than training stability.

8.3 Results

Metric	Baseline GAN	Spectral-Norm GAN
G-loss (final)	~ 2.28	~ 0.71
D-loss (final)	~ 0.66	~ 1.37
G-loss variance	0.112769	0.000011
Variance reduction	-	100.0%
Detection AUC	1.000	1.000
Efficacy	0.861	0.992

All three predictions were confirmed. G-loss variance was reduced by 100%, from 0.113 to 0.000011. The final G-loss of 0.71 is very close to $\ln(2) = 0.693$, the generator loss at the true Nash equilibrium (G perfectly matches p_{data} , D outputs 0.5 everywhere). The D-loss of 1.37 is equally diagnostic: with two-sided label smoothing (real target 0.9, fake target 0.1), the theoretical D-loss when D outputs 0.5 for all samples is $\text{BCE}(0.5, 0.9) + \text{BCE}(0.5, 0.1) = 0.693 + 0.693 \approx 1.386$ - exactly matching the observed 1.37, confirming that SN drove training to the true Nash equilibrium rather than the discriminator-dominance plateau (D-loss ≈ 0.65) seen in the baseline. Efficacy improved from 0.861 to 0.992, a gain of 13 percentage points, indicating the generator learned a much more faithful representation of the data distribution when its gradient signal was stabilised. Loss curves are shown in Appendix Fig. A21.

Prediction (3) - detection - was not confirmed: AUC remained at 1.000 even at the Nash equilibrium. This negative result is itself informative. The SN experiment decouples two failure modes that are conflated in the baseline: training instability (gradient saturation, discriminator dominance) and model capacity (the MLP's ability to represent the joint distribution). SN eliminates the first completely - confirmed by the Nash equilibrium loss values - yet detection AUC does not move. This isolates model capacity as the binding constraint: a 3-layer MLP generator cannot reproduce the high-dimensional tabular distribution regardless of how well-trained it is. Improving detection AUC would require a more expressive architecture, not better training stability.

Appendix: Figures

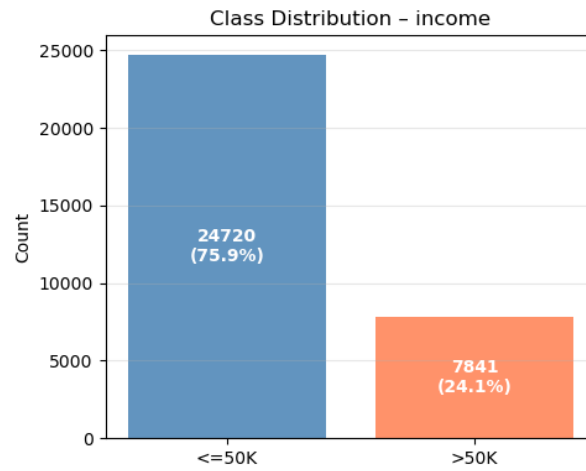


Fig. A1. Class distribution of the income target. The 3:1 imbalance (75.9% vs 24.1%) is preserved by stratified splitting.

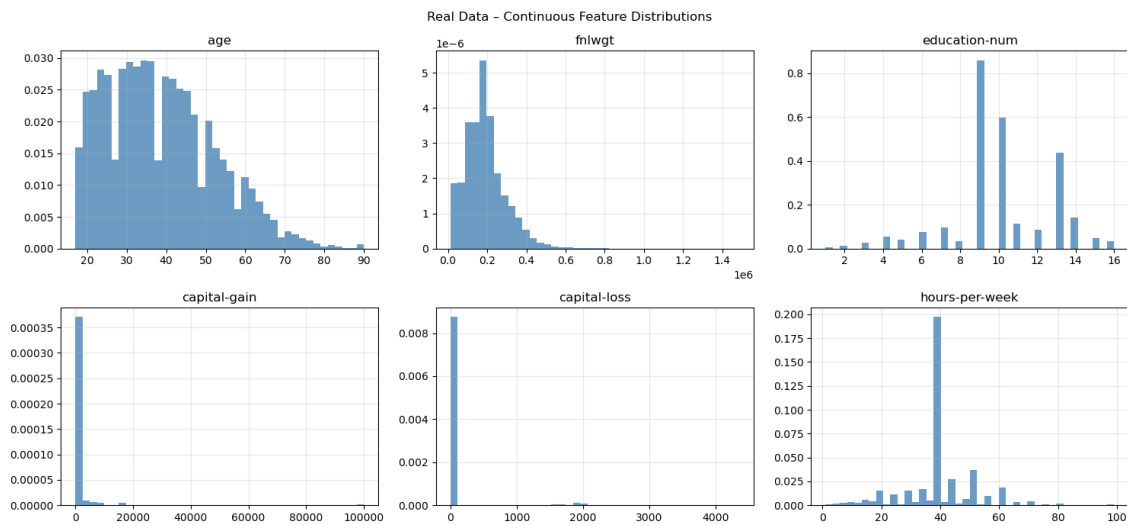


Fig. A2. Continuous feature distributions in the real training data. Capital-gain and capital-loss exhibit heavy positive skew with most values at zero; age and hours-per-week follow roughly bell-shaped distributions.

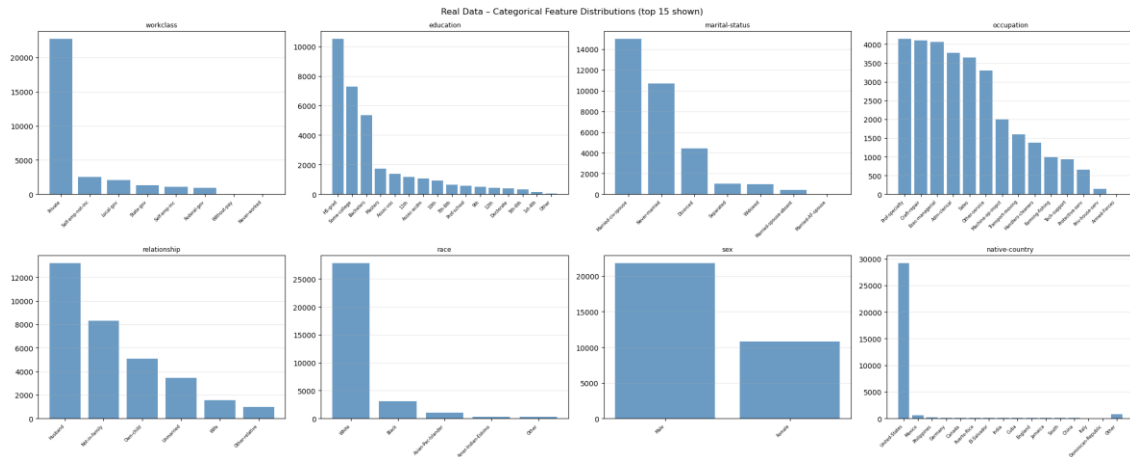


Fig. A3. Categorical feature distributions (top 15 categories shown per feature; remaining values aggregated as "Other"). Native-country is heavily dominated by United-States; occupation and education show broader spread that a GAN must capture to avoid mode collapse.

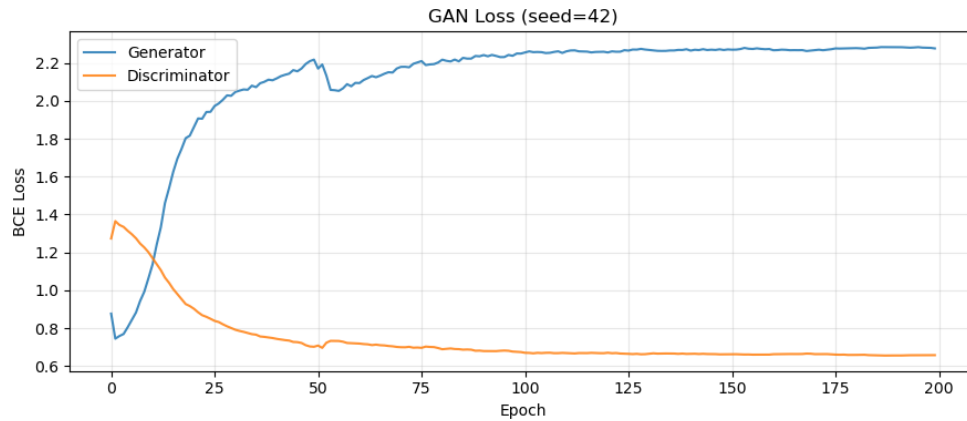


Fig. A4. GAN training loss, seed 42. G-loss climbs from ~2.0 and plateaus near 2.27; D-loss settles near 0.66. The gap between G-loss and $\ln(2)=0.693$ indicates persistent discriminator dominance.

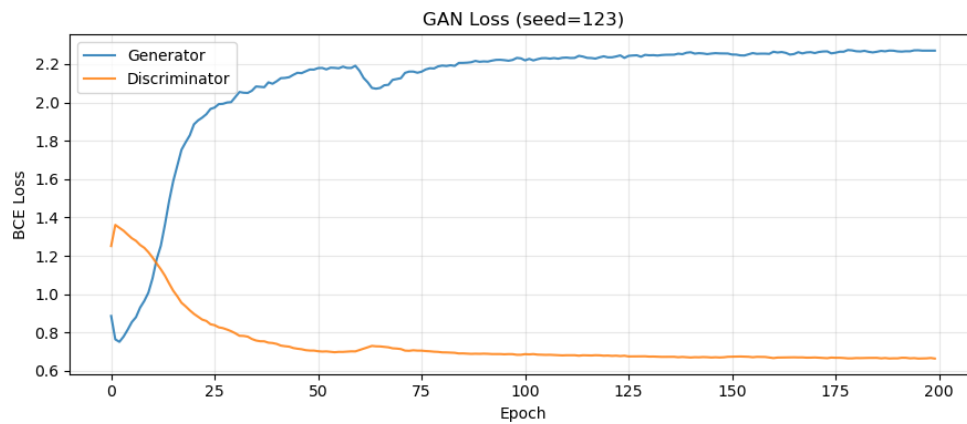


Fig. A5. GAN training loss, seed 123. Pattern is consistent with seed 42: D settles near 0.65, G never recovers to the Nash equilibrium value of 0.693.

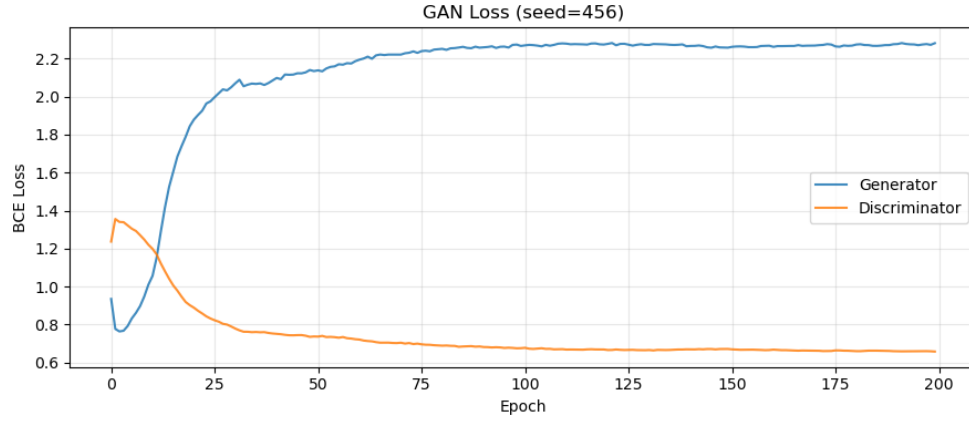


Fig. A6. GAN training loss, seed 456. Consistent D-dominant plateau across all three seeds confirms this is a structural property of the training setup, not seed-specific variance.

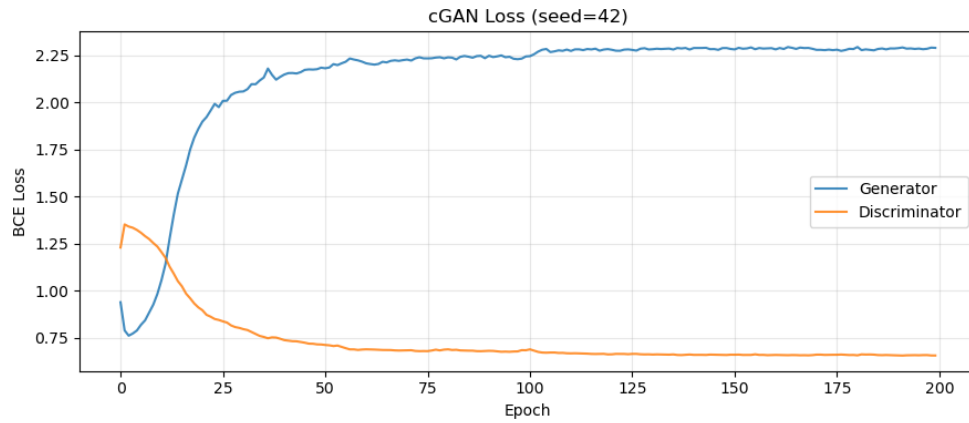


Fig. A7. cGAN training loss, seed 42. Conditioning on the income label does not substantially change the loss dynamics; G-loss again plateaus well above 0.693.

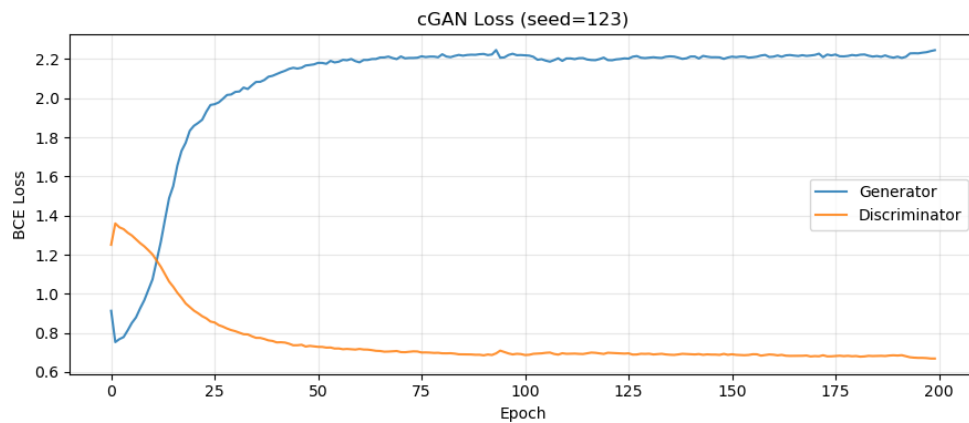


Fig. A8. cGAN training loss, seed 123. G/D pattern mirrors the unconditional GAN; the discriminator retains an advantage regardless of the conditioning signal.

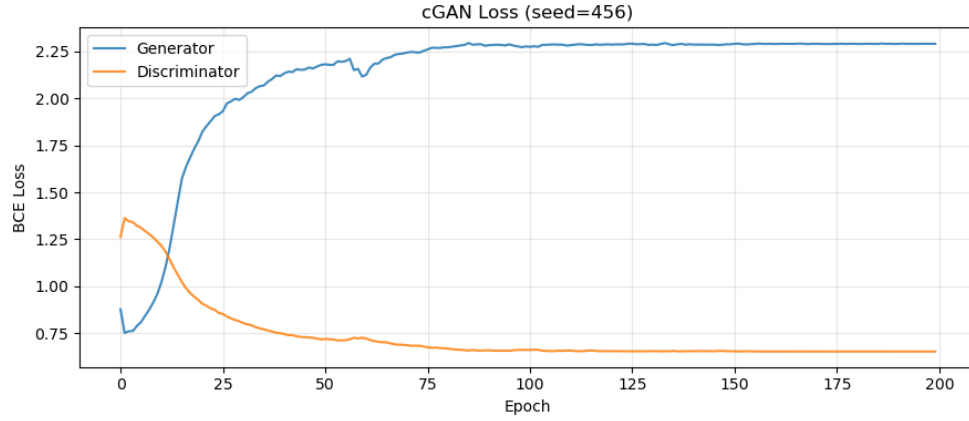


Fig. A9. cGAN training loss, seed 456. Stable convergence with no catastrophic oscillation, but the D-dominant imbalance persists across all seeds.

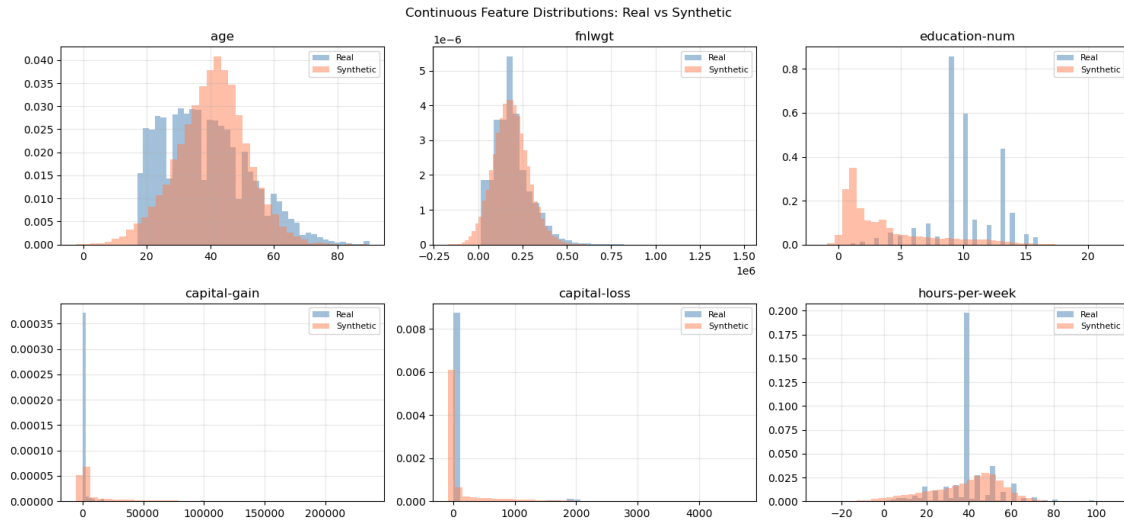


Fig. A10. GAN continuous feature distributions, real vs synthetic (seed 42). Overall shapes are broadly matched; capital-gain and capital-loss heavy tails are underrepresented in the synthetic data.

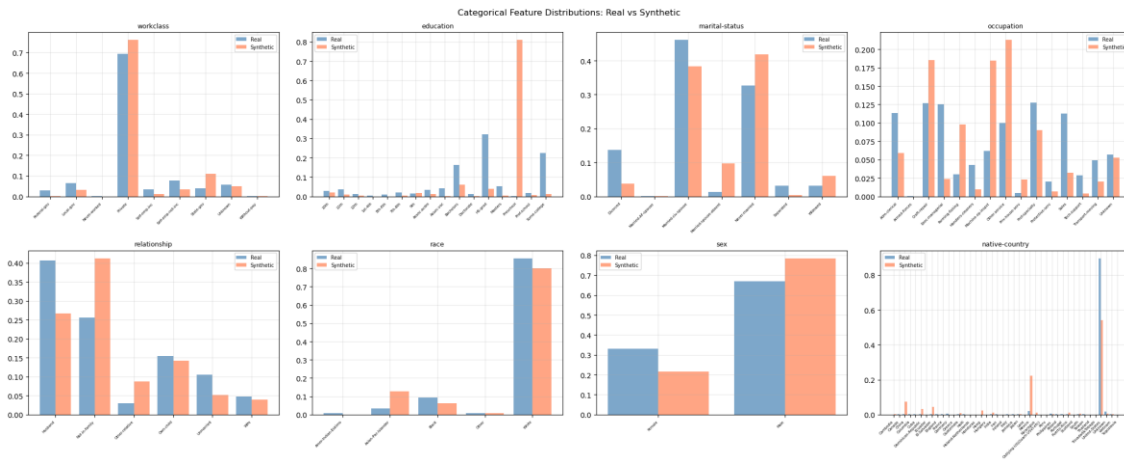


Fig. A11. GAN categorical feature distributions, real vs synthetic (seed 42). High-frequency categories are roughly reproduced; low-frequency categories (e.g., rare native-country values) are systematically under-represented.

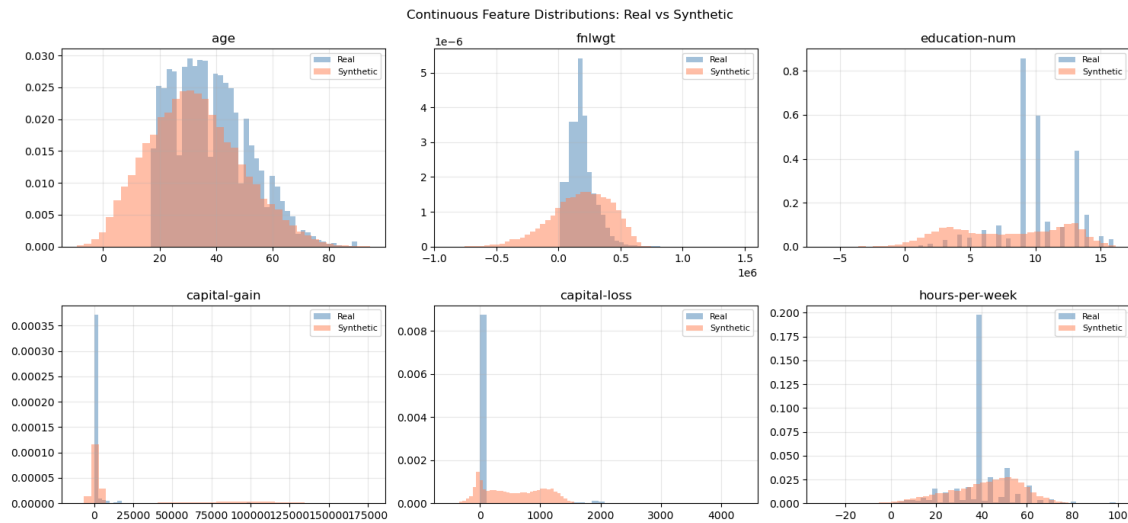


Fig. A12. cGAN continuous feature distributions, real vs synthetic (seed 42). Conditional generation produces similar continuous-feature coverage to the unconditional GAN; heavy-tail features remain challenging.

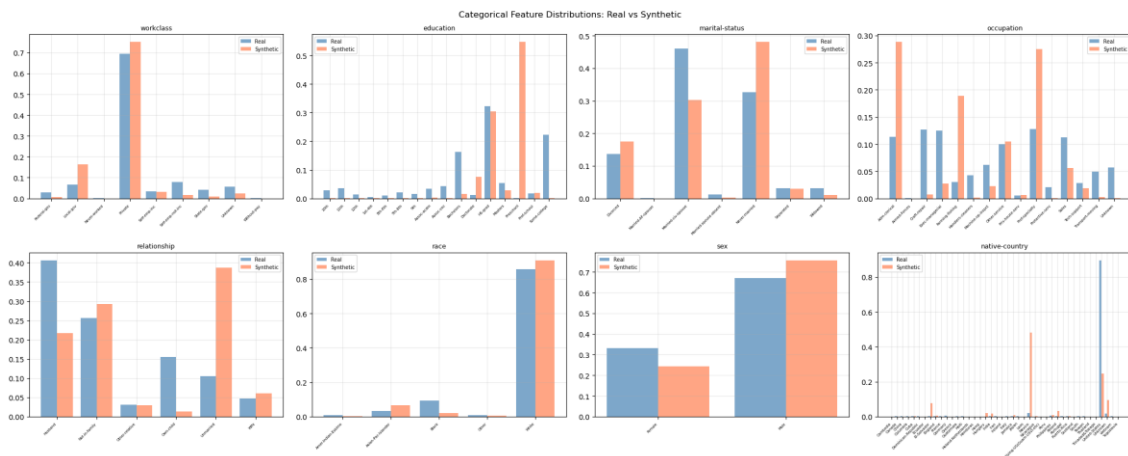


Fig. A13. cGAN categorical feature distributions, real vs synthetic (seed 42). Class-conditional generation preserves category proportions somewhat better than the unconditional GAN for occupation and marital-status.

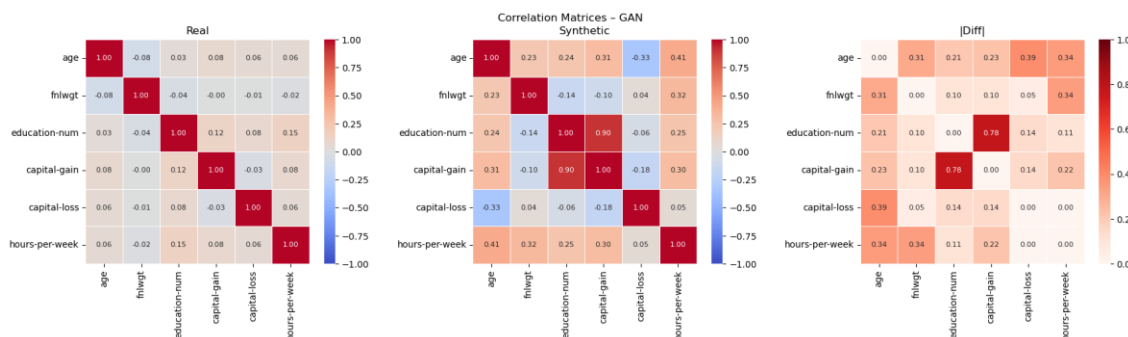


Fig. A14. GAN correlation matrices: real (left), synthetic (center), difference (right). Non-trivial residuals in the difference matrix indicate the generator captures only first-order marginals and misses higher-order inter-feature dependencies.

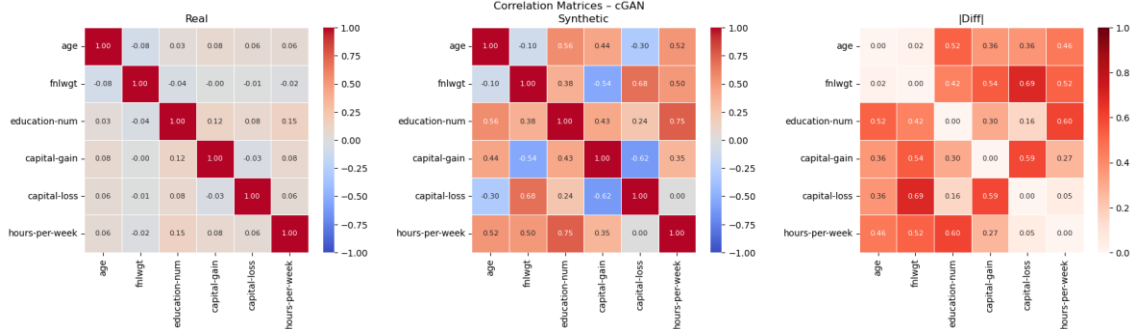


Fig. A15. cGAN correlation matrices: real (left), synthetic (center), difference (right). Residuals are comparable to the unconditional GAN; conditioning does not substantially improve correlation structure reproduction.

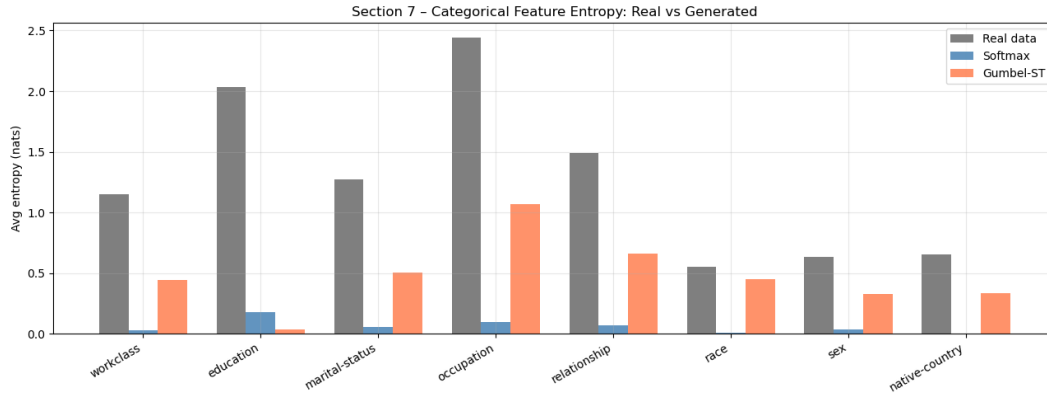


Fig. A16. Per-feature Shannon entropy: real data vs softmax vs Gumbel-ST. Softmax entropy collapses near zero across all features (near-uniform soft distributions); Gumbel-ST entropy closely tracks the real-data reference.

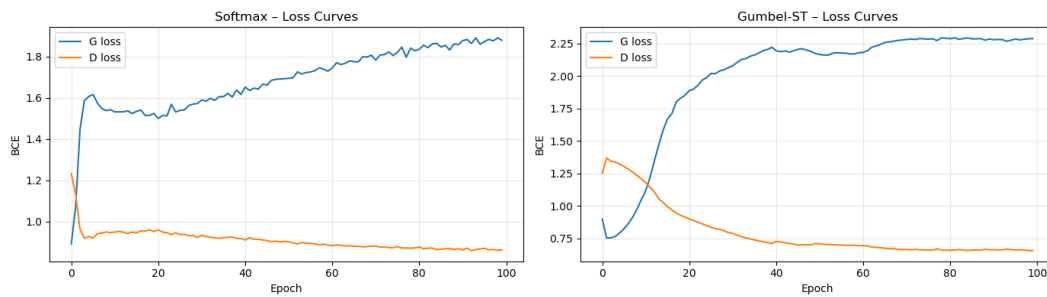


Fig. A17. Training loss comparison: softmax vs Gumbel-ST. Gumbel-ST reaches a lower D-loss (~0.66) than Softmax (~0.86), converging close to the discriminator-dominance optimum (~0.65) because D must discriminate on genuine content; Softmax D-loss is higher because the format-mismatch signal creates a degenerate training landscape.

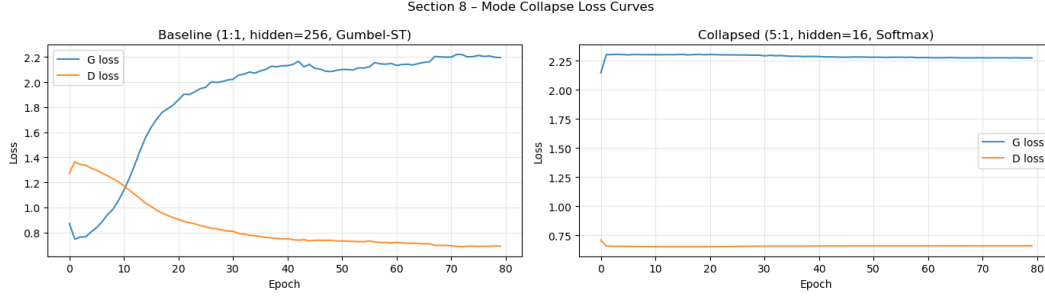


Fig. A18. Section 8 loss curves: baseline (1:1, hidden=256, Gumbel-ST) vs collapsed (5:1 D:G, hidden=16, softmax). Both runs reach similar G and D plateaus (~2.27 and ~0.66 respectively), confirming that mode collapse is not visible in scalar losses - only the coverage metric (0.116 vs 1.031) reveals the narrowing of the output distribution.

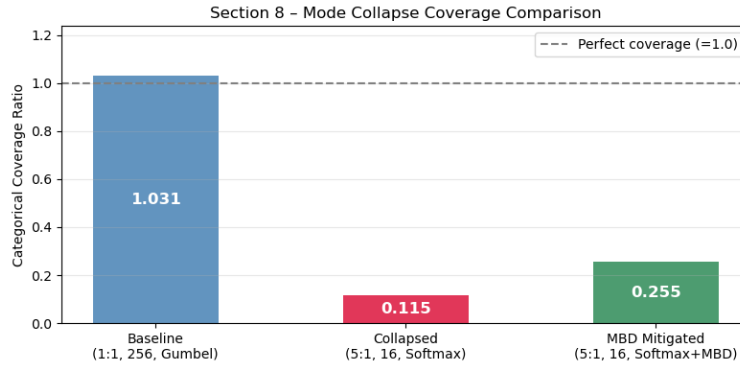


Fig. A19. Categorical coverage ratios: baseline (1.031), collapsed (0.116), and MBD-mitigated (0.255). The collapsed run produces only 44 out of 381 real unique combinations; MBD more than doubles diversity to 97 combinations, with the capacity bottleneck preventing full recovery to baseline.

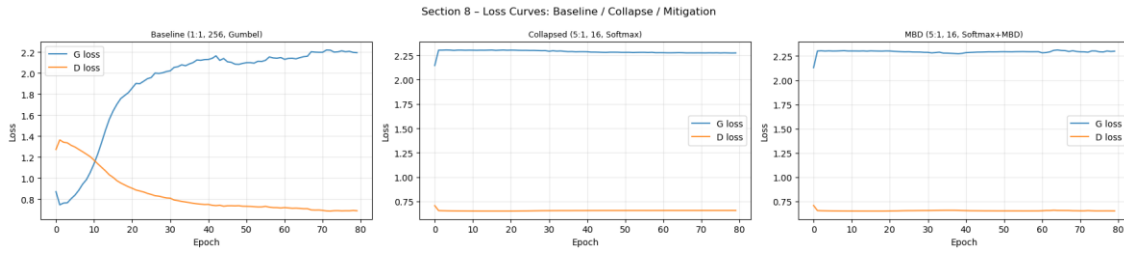


Fig. A20. Section 8 loss curves: all three configurations (baseline / collapsed / MBD). All three settle at similar G/D loss values (~2.27-2.30 / 0.65-0.69), confirming that the coverage difference is driven by architectural choices (capacity bottleneck, Gumbel noise removal) rather than loss-curve-visible training dynamics.

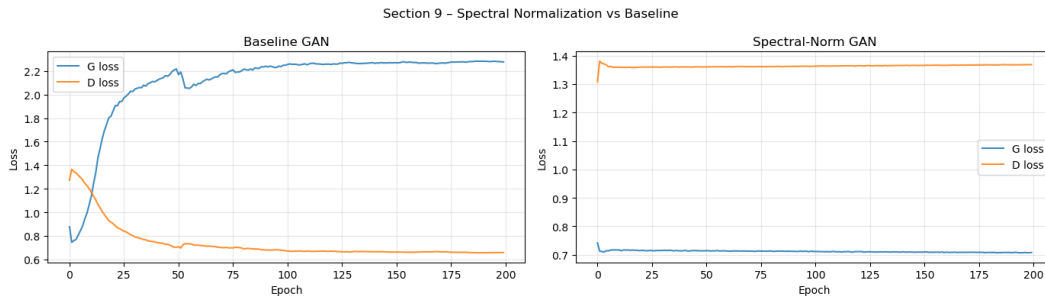


Fig. A21. Section 9 loss curves: baseline vs spectral-norm GAN. Spectral-norm G-loss stabilizes near $\ln(2)=0.693$ (the Nash equilibrium) while D-loss rises to ~ 1.36 , a stark contrast to the baseline D-dominant plateau.